

Competing *de dicto* terms reduce *de re* acceptability

Yuhan Zhang & Kathryn Davidson

yuz551@g.harvard.edu, Chicago Linguistic Society, April 24, 2022



Department of Linguistics
Harvard University

Meaning & Modality (M&M) Lab

Introduction: intensionality

- Words or phrases like *believe, want, may, it is possible* ... enable us to express things that are not about the actual here and now.
- They are the so-called intensional operators that add more expressivity to language to convey information about different possibilities.
- Yet they also give rise to a certain ambiguity.

Attitude verb + Noun phrase ➡ Ambiguity !

- Aurora wants to marry a prince.



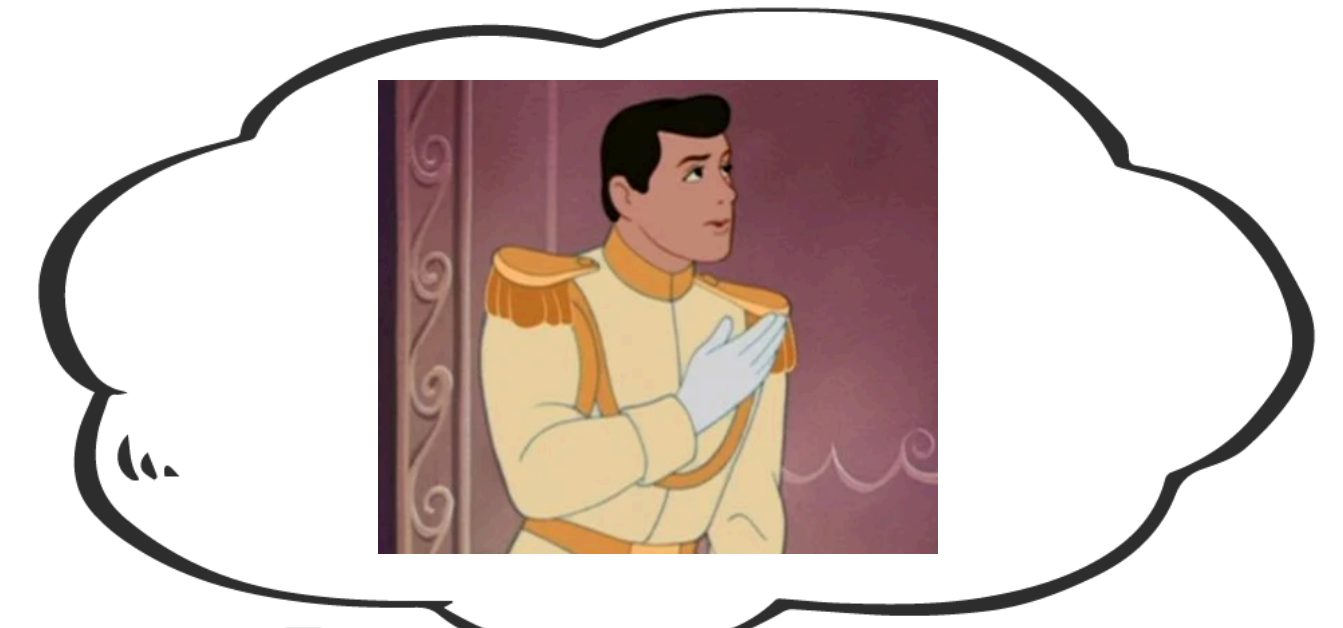
There is a specific person in the actual world who is a prince and Aurora wants to marry him.

de re
(About the thing)



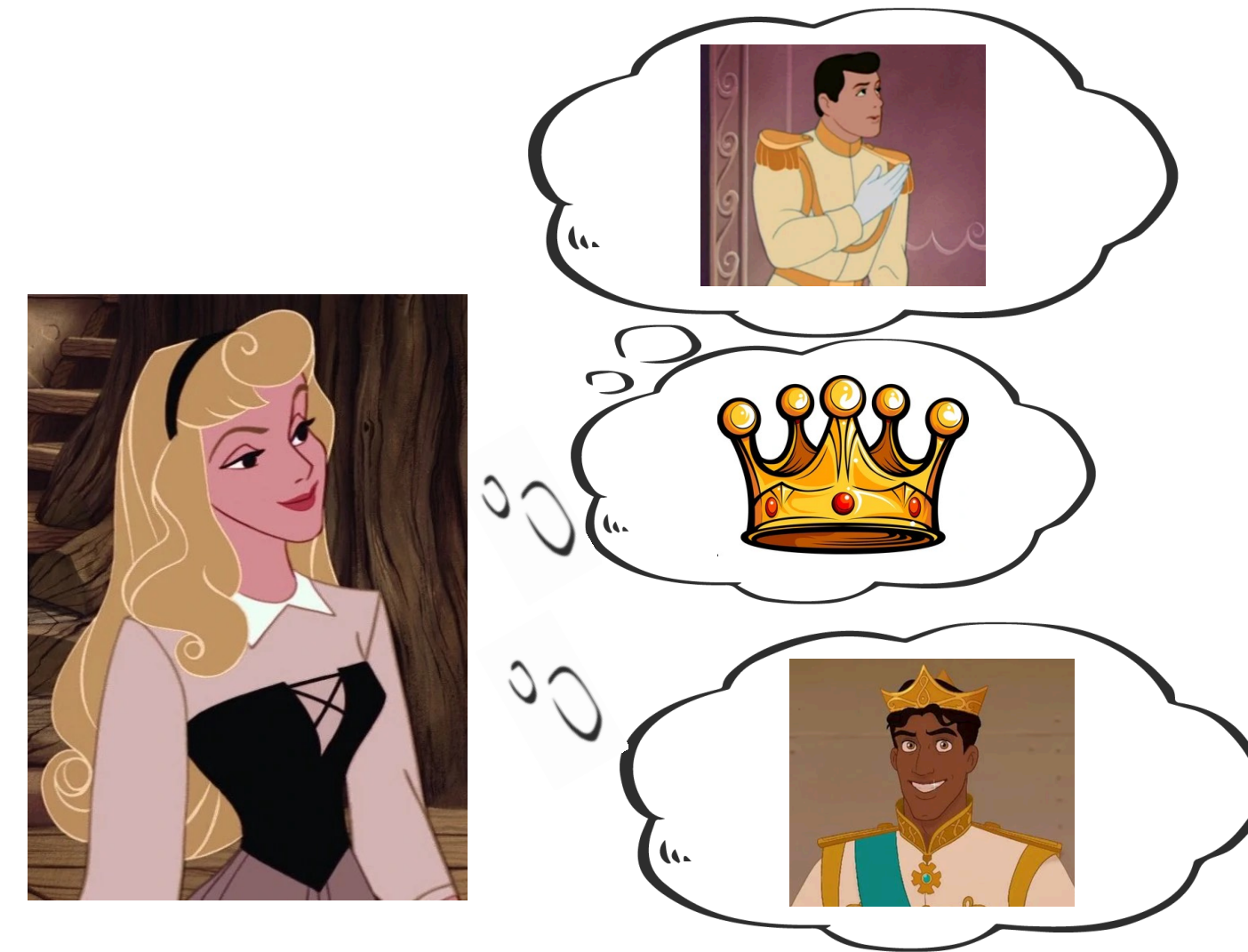
Aurora holds the wish of “marrying a prince” although she has no particular one in mind.

de dicto
(About the statement or the word)



De re/de dicto in indefinite NPs

- In formal semantics, the most classic way to capture this ambiguity is using the scope interaction between the existential quantifier a and the attitude verb.


$$\exists x [\text{prince}(x) \wedge \forall w [\text{WANT}(A, w) \rightarrow \text{marry}(A, x)]]$$

de re

$$\forall w [\text{WANT}(A, w) \rightarrow \exists x [\text{prince}(x) \wedge \text{marry}(A, x)]]$$

de dicto

Implication from the formal analysis

$$\exists x [\text{prince}(x) \wedge \forall w [\text{WANT}(A, w) \rightarrow \text{marry}(A, x)]]$$

de re

$$\forall w [\text{WANT}(A, w) \rightarrow \exists x [\text{prince}(x) \wedge \text{marry}(A, x)]]$$

de dicto

- For the context that allows the *de re* reading, Aurora doesn't have to know that the person who she wants to marry is a prince.
 - The term “a prince” is just a convenient referential term that picks up the individual in the real world.
- For the context that allows the *de dicto* reading, Aurora knows that whoever she wants to marry is a prince.
 - Aurora commits to the fact that if there is a person in her mind she wants to marry, that person is “a prince”.

Attitude verb + Indefinite noun phrase

➔ **Ambiguity**

- Semantists have come up with various kinds of mathematical tools to capture this intricate semantic ambiguity (e.g., Charlow & Sharvit, 2014; Keshet, 2011; Percus, 2010).
- However, things get a little unclear when attitude verbs combine with other kinds of noun phrases.

Attitude verb + Definite noun phrase

? ? **Ambiguity**

Contradictory linguistic judgment of definite *de re* NPs

De re condition: John reviewed an abstract which is anonymous to him. He told me that it will be accepted. Judging from the abstract content, I know that this abstract is written by you. To inform John's opinion to you, I utter the sentence above.

(1) John believes that your abstract will be accepted. (von Fintel & Heim, 2011)

True in von Fintel & Heim's view

De re condition: John wants to meet a certain boy, a boy who happens, as a matter of fact, to be the one boy who failed the exam. John doesn't recognize the description "the boy who failed the exam" as a description of the boy he has in mind.

(2) John wants to talk to the boy who failed the exam. (Fodor, 1970)

False in Fodor's view



John doesn't recognize the boy under the description "the boy who failed the exam".

Contradictory linguistic judgment of definite *de re* NPs

De re condition: Sally is watching a person across the room. He is laughing and happens to be Sally's brother, although she doesn't realize he is her brother. Sally says to herself, "He [pointing at the laughing man] is happy."

(3) Sally believes that her brother is happy. (Nelson, 2022)

False in *Stanford Encyclopedia of Philosophy*



Sally doesn't conceptualize the man as her brother.

De re readings for definite noun phrases in belief reports **should be true** given the corresponding context.

von Fintel & Heim (2011)

Romoli & Sudo (2009)

Naive Russellians



De re readings for definite noun phrases in belief reports **should be false** given the corresponding context.

Nelson (2022)

Fodor (1970)

Partee et al. (1990)

Gottlob Frege

Research questions

- We seem to take for granted that the ambiguity caused by the interaction of intensional operators and noun phrases can be generalized to lots of cases.
- The contradictory linguistic judgments for *de re* readings of definite noun phrases question the generalization.
- Are *de re* readings of definite noun phrases true or false (acceptable or not)?
- In other words, does the *de re/de dicto* ambiguity hold for definite noun phrases like *your abstract*, *the boy who failed the exam*, etc?
- If the *de re* reading is indeed not acceptable, is it a grammar design or a pragmatic artifact?
- What linguistic or contextual factors affect the availability of certain ambiguous readings?

Experiment 1

- **Question: Do native speakers of English accept *de re* reading of definite NPs?**
- **Paradigm: sentence evaluation task given a context, same sentence different contexts**

de re reading

Context: Julie is a judge of an ongoing poetry competition. She encounters an extremely well-written poem and believes that this poem will be the winner of the competition. This poem happens to be written by Elizabeth Johnson, a well-known poet in the US. But unfortunately, as a judge, Julie is blind to the authors and therefore does not know it is Elizabeth Johnson who wrote this excellent poem.

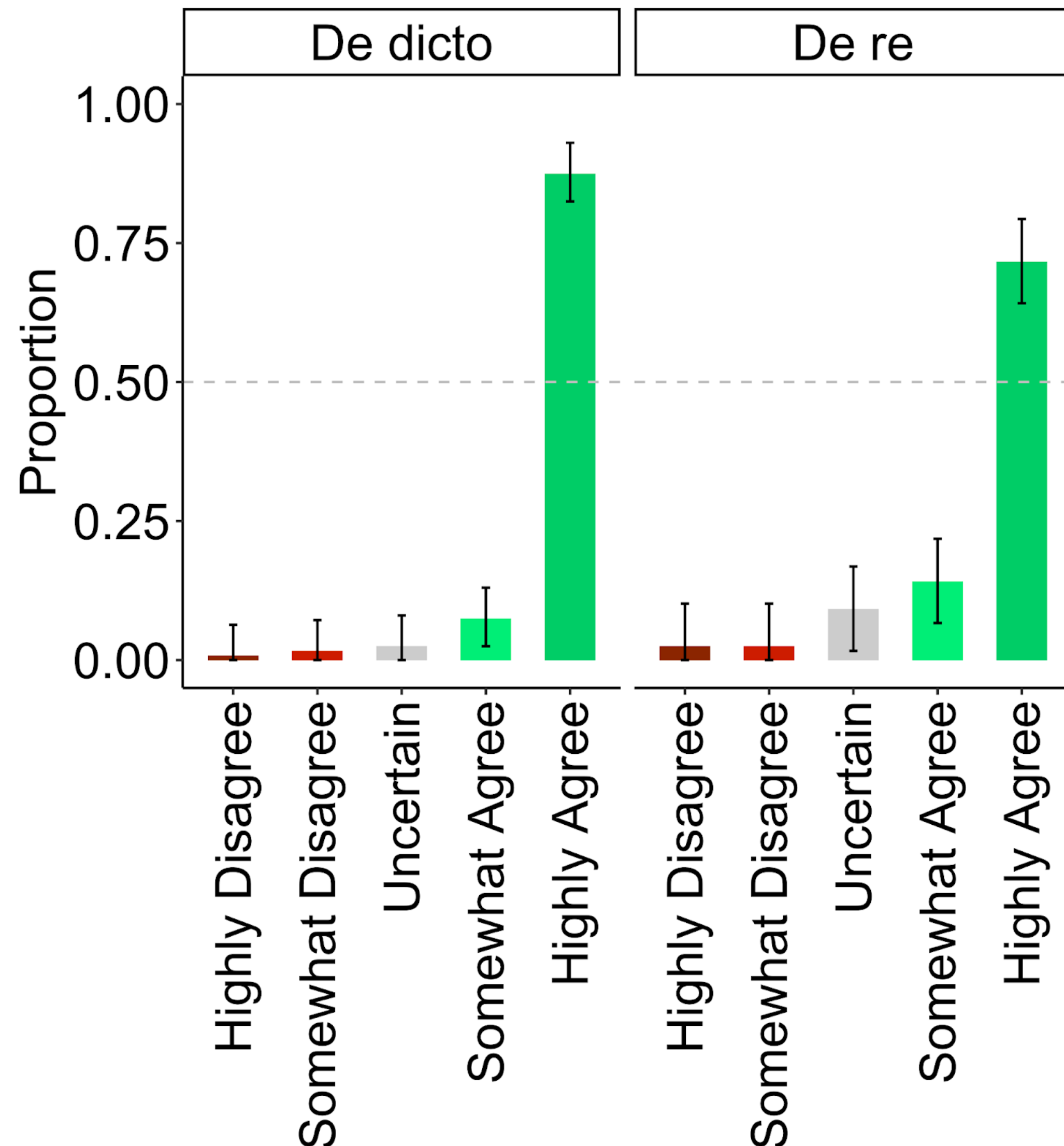
de dicto reading

Context: Julie is a judge of an ongoing poetry competition. She is told that Elizabeth Johnson, one of the best-known poets in the US, also submitted a poem to the competition. Julie is a huge fan of Elizabeth. Even though Julie is blind to the authors and does not know which poem is written by Elizabeth, she believes that no matter which poem Elizabeth submitted, it will win the competition.

According to the context, do you agree or disagree with the following belief report?

Belief report to be evaluated: Julie believes that Elizabeth's poem will win the competition.

Experiment 1



- Methods
- Participants (N = 66), four context items
- Within-participants design on the context condition
- Results
 - The majority of participants accepted the *de re* interpretation of definite noun phrases.
 - Although statistically, the *de re* reading is less likely to be accepted compared with the *de dicto* interpretation ($\beta = -0.553$, $SE = 0.179$, $p = .002$).
- Discussion
 - Even if we regard “somewhat agree” and options on its left as tending to disagreeing with *de re*, the *de re* reading is not as bad as the literature has argued.

Experiment 2

- **Question:** If one single context theoretically allows two readings, is the *de re* reading acceptable?
- **Paradigm:** sentence evaluation task given a context, same context with two sentences

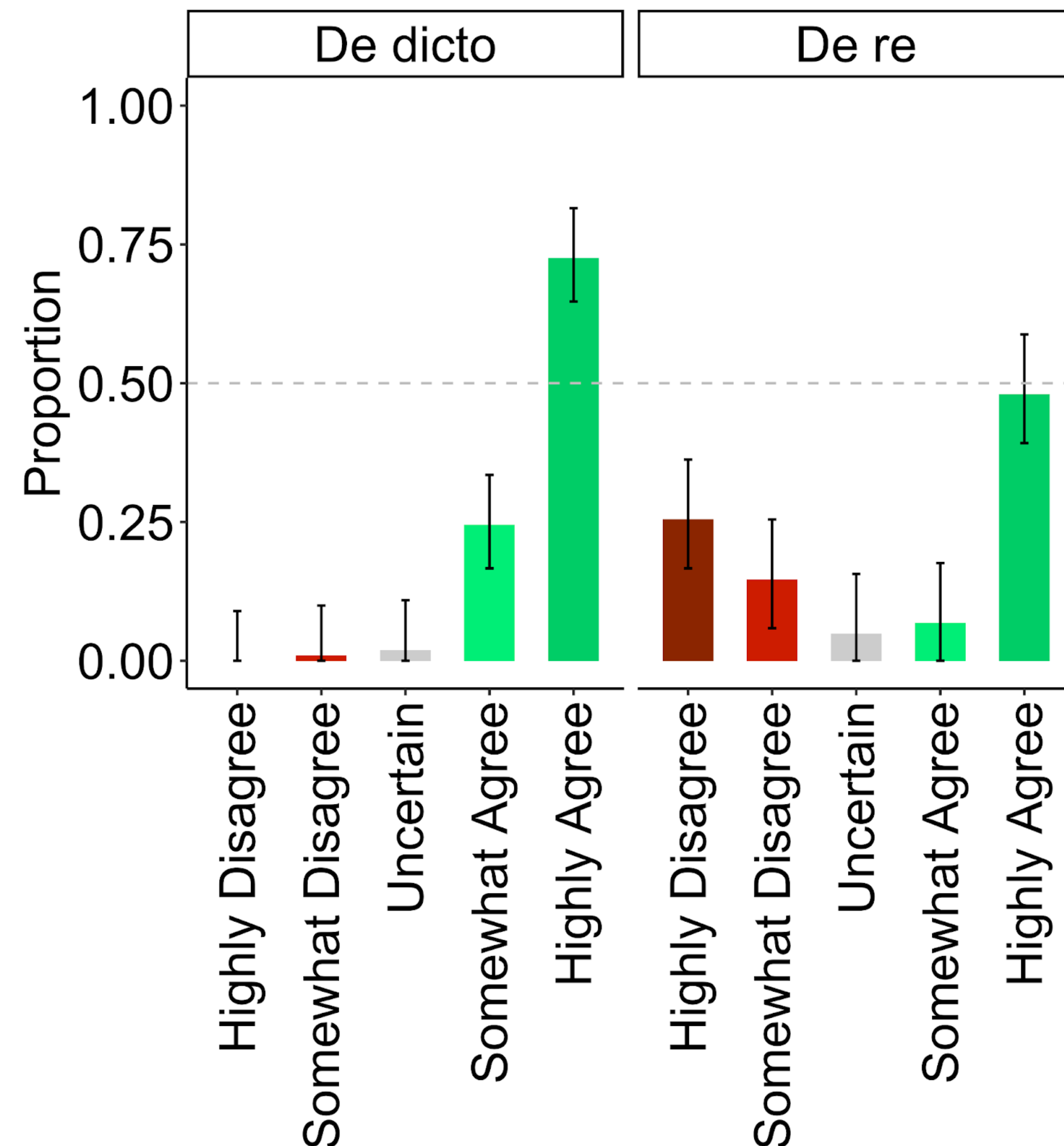
Context: Julie is one of the judges of an ongoing poetry competition. The best poem that she has read so far is an extremely intriguing poem about the ocean. She believes that this poem will win the competition. Julie remembers being told that Nicole, one of the best-known poets, submitted a poem about the ocean to the competition. Therefore, Julie concludes that this poem must be written by Nicole and the first prize will be going to her. However, this poem was actually written by Elizabeth, a younger and lesser-known poet. It is just a coincidence that the two poets wrote about the same topic.

Participants were randomly assigned to one of the two sentences:

De dicto condition: Julie believes that Nicole's poem will win the competition.

De re condition: Julie believes that Elizabeth's poem will win the competition.

Experiment 2



- Methods
 - Participants (N = 60) and four context items
 - Within-participants design on sentence condition
- Results
 - The *de dicto* reading was highly agreed with.
 - The *de re* reading received bimodal distribution (Zhang & Davidson, 2021).
- Discussion
 - *De re* is clearly disliked in this setup
 - There is still a large proportion of “highly agree” with *de re*.

Interim summary

- In Experiment 1, we found that *de re* and *de dicto* readings are acceptable in contexts that theoretically bias only one reading.
- In Experiment 2, we found that in contexts that theoretically allow both *de re* and *de dicto* readings, belief reports that have the *de dicto* reading are preferred, although it is not the case that *de re* readings are disliked at all.
- Proposal
 - The competing *de dicto* term drives the bimodal distribution of *de re*.
- Experiment 3
 - Judge a *de re* sentence in (a) the context that allows both a *de re* and a *de dicto* way to refer to an object; (b) the context that only biases the *de re* reading.
 - Predict that *de re* will only show bimodal distribution in context (a).

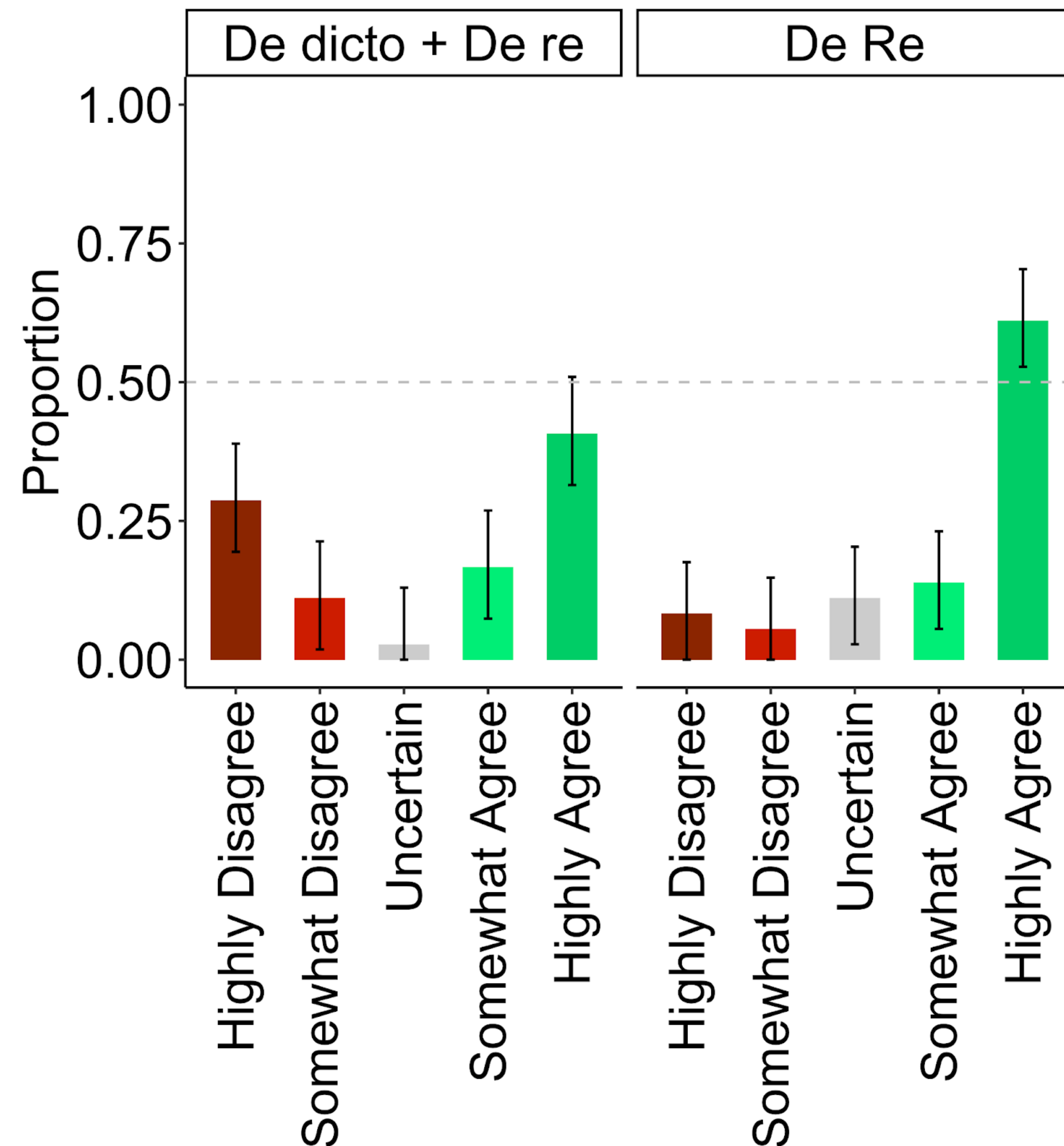
Experiment 3

De dicto + de re context: Julie is one of the judges of an ongoing poetry competition. The best poem that she has read so far is an extremely intriguing poem about the ocean. She believes that this poem will win the competition. Julie remembers being told that Nicole, one of the best-known poets, submitted a poem about the ocean to the competition. Therefore, Julie concludes that this poem must be written by Nicole and the first prize will be going to her. However, this poem was actually written by Elizabeth, a younger and lesser-known poet. It is just a coincidence that the two poets wrote about the same topic.

Control Context: Julie is a judge of an ongoing poetry competition. She encounters an extremely well-written poem and believes that this poem will be the winner of the competition. This poem happens to be written by Elizabeth Johnson, a well-known poet in the US. But unfortunately, as a judge, Julie is blind to the authors and therefore does not know it is Elizabeth Johnson who wrote this excellent poem.

Belief report to be evaluated: Julie believes that Elizabeth's poem will win the competition.

Experiment 3



- Methods
 - Participants (N = 60), four context items
 - Within-participants design on context condition
- Results
 - *De re* only shows the bimodal distribution when the *de dicto* reading is competing in the same context.
 - Otherwise, *de re* is highly agreed upon.
- Discussion
 - Our hypothesis is confirmed.

Conclusion

- With three experiments, we find that *de re* readings of definite noun phrases in belief reports are acceptable (e.g., von Stechow & Heim, 2011; Romoli & Sudo, 2009)
- However, the acceptability of *de re* is affected by whether a competing *de dicto* reading also exists in the same context.
- We conclude that the *de re* interpretations of definite noun phrases are in the grammar of English but its acceptability might be affected by contextual factors in the pragmatic region.

Discussion

- How to account for Fodor's judgment and the view from *SEP*?
- The interaction between formal semantics and experiments
- Future directions
 - Linguistics and philosophy
 - Propositional attitude report
 - Frege, Naive Russellian, Neo-Russellian
 - The acceptability of *de re/de dicto* readings for different attitude verbs, noun phrase constructions, and different languages.
 - Other contextual/pragmatic factors that influence the acceptability of *de re*.

Acknowledgement

- Special thanks to members at the Meaning & Modality Lab.
- Reviewers and audience at CLS.



Thank you!

Selected references

- Charlow, S., & Sharvit, Y. (2014). Bound “de re” pronouns and the LFs of attitude reports. *Semantics and Pragmatics*, 7(3), 1–43. <https://doi.org/10.3765/sp.7.3>
- Fodor, J. D. (1970). *The Linguistic Description of Opaque Contexts*. Massachusetts Institute of Technology.
- Frege, G. (1892). Sense and Reference. *THE PHILOSOPHICAL REVIEW*, 209–230.
- Keshet, E. (2011). Split intensionality: A new scope theory of de re and de dicto. *Linguistics and Philosophy*, 33(4), 251–283. <https://doi.org/10.1007/s10988-011-9081-x>
- Partee, B. H., Meulen, A. ter, & Wall, R. E. (1990). *Mathematical methods in linguistics*. Kluwer Academic.
- Quine, W. V. (1956). Quantifiers and Propositional Attitudes. *The Journal of Philosophy*, 53(5), 177. <https://doi.org/10.2307/2022451>
- Romoli, J., & Sudo, Y. (2009). De Re/De Dicto Ambiguity and Presupposition Projection. *Proceedings of Sinn Und Bedeutung*, 13(2), 14.
- Russell, B. (1905). On Denoting. *Mind*, 14(56), 479–493.
- von Fintel, K., & Heim, I. (2011). *Intensional Semantics*.
- Zhang, Y., & Davidson, K. (2021). De re interpretation in belief reports: An experimental investigation. *Experiments in Linguistic Meaning*, 1, 310-321.

Terminologies

Possible meanings	This presentation	Fodor (1970)	Jaszczolt (1997)	Description
Base	—	—	—	a belief about a particular thing; the NP refers to the thing both in reality and in the BH's mind.
Referential, reality-oriented	<i>de re</i>	transparent	<i>de re</i>	a belief about a particular thing; the NP refers to the thing only in reality, not in BH's mind.
Referential, BH-oriented	<i>de dicto</i>	—	<i>de dicto-1</i>	a belief about a particular thing; the NP refers to the thing only in BH's mind but not in reality.
Non-referential, attributive	<i>de dicto</i>	opaque	<i>de dicto-2</i>	a belief with respect to a particular name, about whoever or whatever undergoes the name or fits the expression.

Transparency/opacity vs *de re/de dicto*

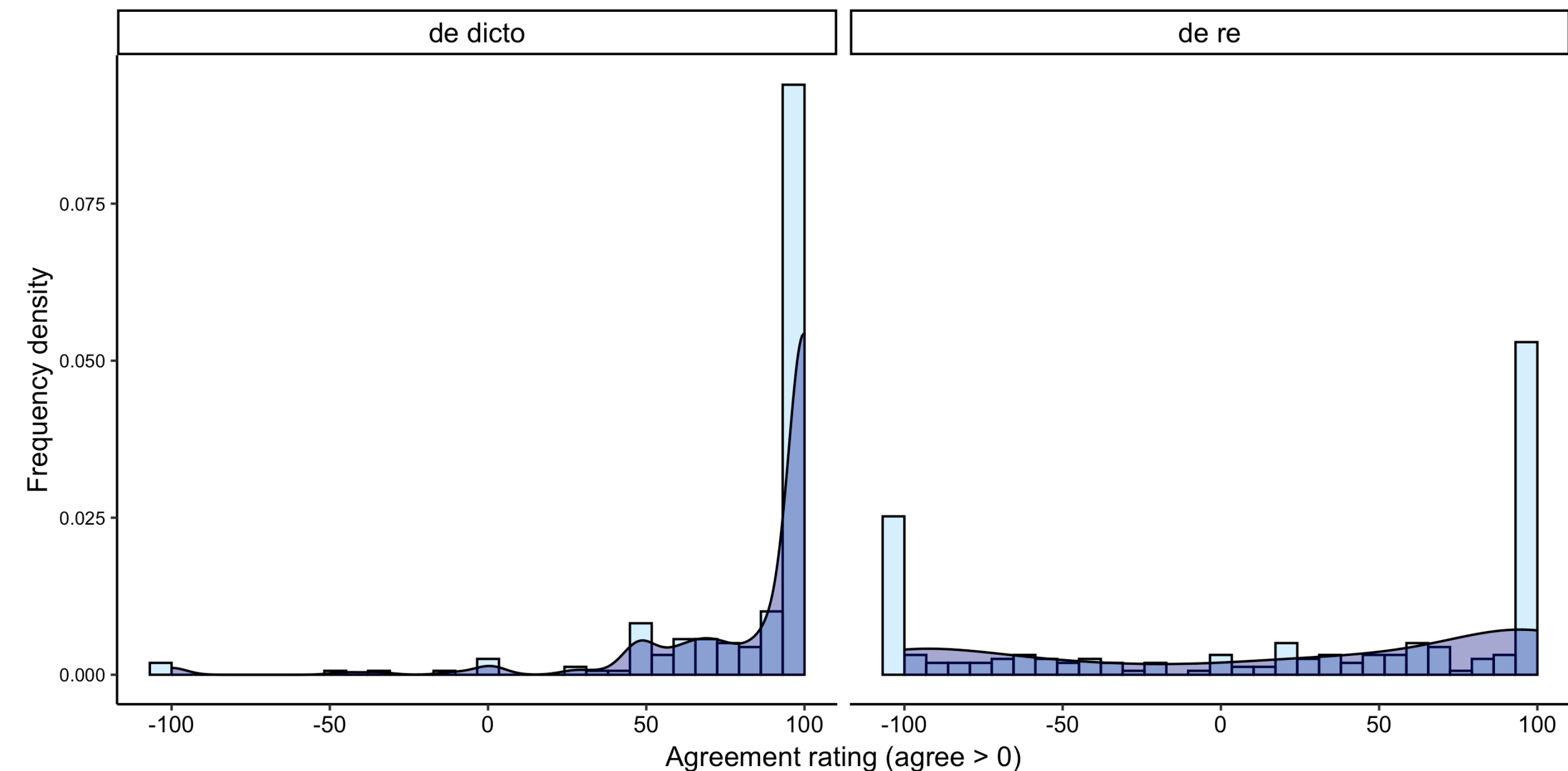
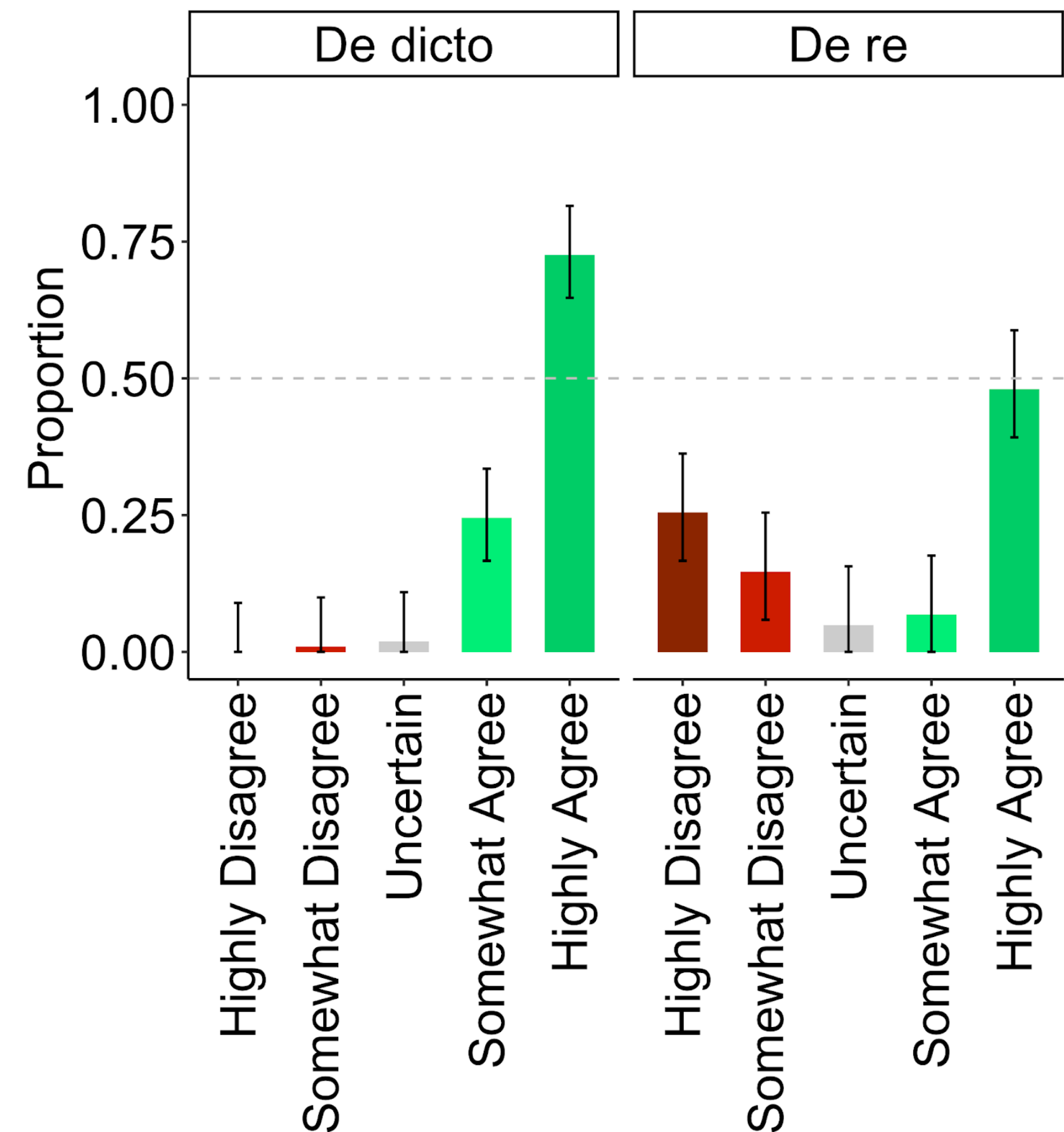
- The definitions of transparency/opacity still need more understanding.
- **Transparency**
 - Transparent sentences allow existential generalization and substitution of coreferential terms (Fodor, 1970).
 - = sentences with no intensional operators.
 - = *de re* reading?
 - = the baseline interpretation?
 - Aurora wants to marry a prince
 - With the reading that there is a specific person Aurora wants to marry; he is a prince; Aurora knows that he is a prince.
 - In a belief report, only the speaker (not the belief holder, i.e., the opaque verb's subject) recognizes the object under the descriptive content of NP.
- **Opacity** (Fodor, 1970)
 - Opaque constructions give rise to the kind of ambiguity discussed in this talk.
 - Opaque contexts give rise to opaque readings.
 - = *de dicto* reading of an ambiguous sentence?
 - In a belief report, both the speaker and the belief holder recognize the object under the descriptive content of NP.

Other theoretical approaches to model *de re/de dicto*

- Situation pronoun (Percus, 2010)
- Concept generator (Charlow & Sharvit, 2014)
- Split intentionality (Keshet, 2008; 2011)
- Exceptional scope (Elliot, 2021)

Likert scale vs. Continuous slider

- Different judgment collection metrics but same distribution pattern



Zhang & Davidson (2021)
Proceedings of Experiment in Linguistic Meaning, 1

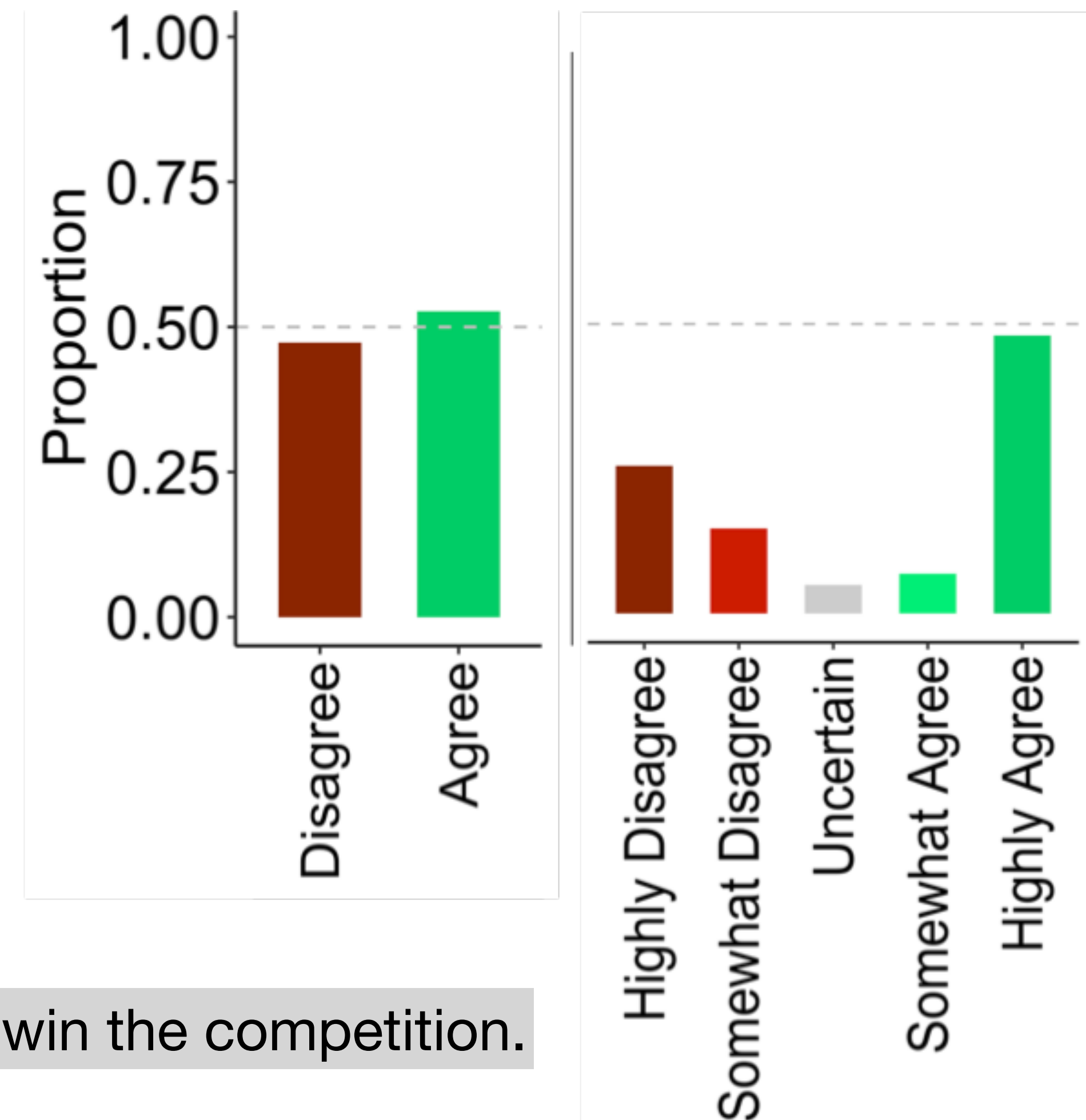
Linking hypothesis

For a thorough discussion about linking hypothesis, please refer to **Jasbi, Waldon & Degen (2019)**, *Frontiers in Psychology*

Experiment 2

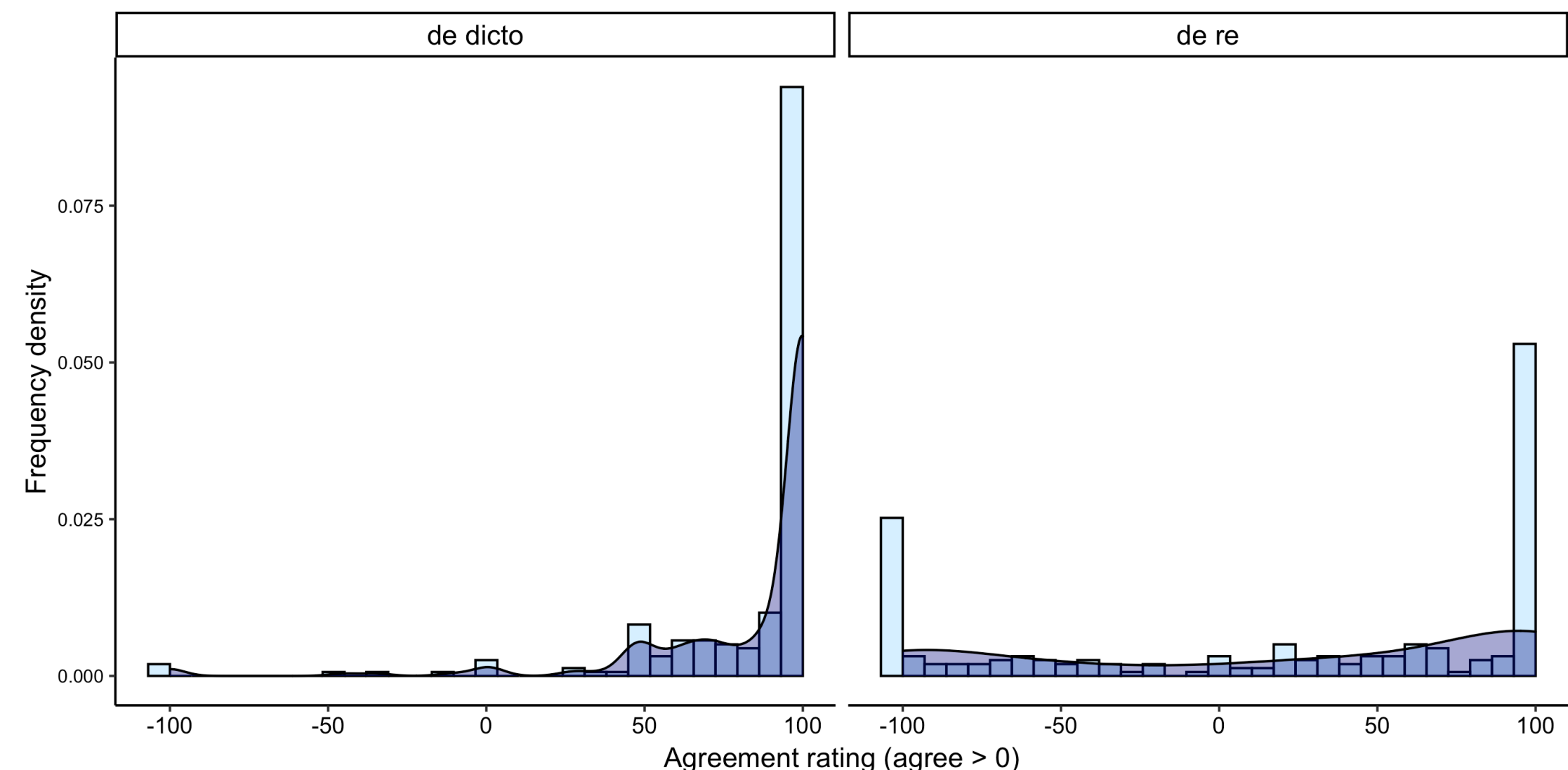
Context: Julie is one of the judges of an ongoing poetry competition. The best poem that she has read so far is an extremely intriguing poem about the ocean. She believes that this poem will win the competition. Julie remembers being told that Nicole, one of the best-known poets, submitted a poem about the ocean to the competition.

Therefore, Julie concludes that this poem must be written by Nicole and the first prize will be going to her. However, this poem was actually written by Elizabeth, a younger and lesser-known poet. It is just a coincidence that the two poets wrote about the same topic.



De re condition: Julie believes that Elizabeth's poem will win the competition.

Between-participant behavior & demographic factor



- Zhang & Davidson (2021)
- *Post hoc* analysis

# of Participants	Agree with 0 trial	Agree with 1 trial	Agree with 2 trials	Total
<i>de re</i>	21 (18.3%)	45 (36.5%)	52 (45.2%)	115
<i>de dicto</i>	0 (0.0%)	7 (6.1%)	108 (93.9%)	115

Table 2: Proportion of participants by the judgment behavior